

The Abstraction Gap Between Mathematics and Physics: The Hidden Compromise in Dimensional Analysis Resolved

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Abstract

In this dataset we bridge the abstraction gap between mathematics and physics by adopting a geometric perspective of trigonometry. We formalize this by addressing the arc length of a circular sector and defining the circular functions axiomatically as direct functional mappings from the physical parameters of a circular sector – namely the radius and the arc length – to its orthogonal projections. Finally, we demonstrate the utility of this framework by modeling simple harmonic motion and uniform circular motion through these geometric functions.

The circular sector and the circular functions

$$s = \lim_{n \rightarrow \infty} 2^n s_n \quad (1a)$$

$$s_n = \sqrt{r^2 + r^2 - 2ra_n} \quad (1b)$$

$$a_n = \frac{r \sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \quad (1c)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (1d)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{r^2 + r^2 - 2ra_n} \quad (2a)$$

$$a_n = \frac{r \sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \quad (2b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (2c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (3a)$$

$$a_n = \frac{r \sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \quad (3b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (3c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (4a)$$

$$a_n = \frac{r \sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \quad (4b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (4c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (5a)$$

$$a_n = r \frac{\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \quad (5b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (5c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (6a)$$

$$a_n = r \sqrt{\frac{r^2 - a_{n-1}^2}{2r^2 - 2ra_{n-1}}} \quad (6b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (6c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (7a)$$

$$a_n = r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r^2 - 2ra_{n-1}}} \quad (7b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (7c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (8a)$$

$$a_n = r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r(r - a_{n-1})}} \quad (8b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (8c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (9a)$$

$$a_n = r \sqrt{\frac{r + a_{n-1}}{2r}} \quad (9b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (9c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (10a)$$

$$a_n = r \sqrt{\frac{2r(r + a_{n-1})}{(2r)^2}} \quad (10b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (10c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (11a)$$

$$a_n = r \sqrt{\frac{2r^2 + 2ra_{n-1}}{(2r)^2}} \quad (11b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (11c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (12a)$$

$$a_n = r \sqrt{\frac{\left(\sqrt{2r^2 + 2ra_{n-1}}\right)^2}{(2r)^2}} \quad (12b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (12c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (13a)$$

$$a_n = r \sqrt{\left(\frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}\right)^2} \quad (13b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (13c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (14a)$$

$$a_n = r \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r} \quad (14b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (14c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (15a)$$

$$a_n = \frac{r \sqrt{2r^2 + 2ra_{n-1}}}{2r} \quad (15b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (15c)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (16a)$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \quad (16b)$$

$$a_0 = x \quad \text{for } -r \leq x \leq r \quad (16c)$$

$$A_{ORS_{n-1}} = \frac{\overline{RS_{n-1}OX_n}}{2} = \frac{\overline{OS_{n-1}RX_{n-1}}}{2} \quad (17)$$

$$\frac{\overline{RS_{n-1}OX_n}}{2} = \frac{\overline{OS_{n-1}RX_{n-1}}}{2} \quad (18)$$

$$\overline{RS_{n-1}OX_n} = \overline{OS_{n-1}RX_{n-1}} \quad (19)$$

$$\frac{\overline{RS_{n-1}OX_n}}{\overline{RS_{n-1}}} = \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}} \quad (20)$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}} \quad (21)$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}} \sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\overline{RS_{n-1}}} \quad (22)$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}} \sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\sqrt{\overline{OR}^2 + \overline{OS_{n-1}}^2 - 2\overline{OS_{n-1}OX_{n-1}}}} \quad (23)$$

$$\overline{RS_n}^2 = \overline{RX_n}^2 + \overline{X_nS_n}^2 \quad (24)$$

$$\overline{RS_n} = \sqrt{\overline{RX_n}^2 + \overline{X_nS_n}^2} \quad (25)$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{X_nS_n}^2} \quad (26)$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + (\overline{OS_n} - \overline{OX_n})^2} \quad (27)$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{OS_n}^2 - 2\overline{OS_nOX_n} + \overline{OX_n}^2} \quad (28)$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 + \overline{OS_n}^2 - 2\overline{OS_nOX_n}} \quad (29)$$

$$\overline{OX_n}^2 + \overline{RX_n}^2 = \overline{OR}^2 \quad (30)$$

$$\overline{OX_n}^2 + \overline{RX_n}^2 - \overline{OX_n}^2 = \overline{OR}^2 - \overline{OX_n}^2 \quad (31)$$

$$\overline{RX_n}^2 = \overline{OR}^2 - \overline{OX_n}^2 \quad (32)$$

$$\overline{RX_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2} \quad (33)$$

$$A = \frac{rs}{2} \quad (34)$$

$$A_0 = \lim_{n \rightarrow \infty} 2^{n-1} A_{ORS_n S_{n-1}} \quad (35)$$

$$A_0 = \lim_{n \rightarrow \infty} 2^{n-1} \frac{\overline{RS_{n-1} OS_n}}{2} \quad (36)$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1} OS_n}}{2} \quad (37)$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} (2^{n-1} \overline{RS_{n-1}}) \overline{OS_n}}{2} \quad (38)$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} \overline{OS_n} (2^{n-1} \overline{RS_{n-1}})}{2} \quad (39)$$

$$A_0 = \frac{\left(\lim_{n \rightarrow \infty} \overline{OS_n} \right) \left(\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}} \right)}{2} \quad (40)$$

$$A_0 = \frac{\overline{OR} \lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}}}{2} \quad (41)$$

$$A_0 = \frac{\overline{OR} \text{arc } RS_0}{2} \quad (42)$$

$$\begin{aligned} \arccos: \{ (r, x) \in \mathbb{R}^2 \mid r > 0, -r \leq x \leq r \} &\rightarrow \mathbb{R} \\ (r, x) &\mapsto s \end{aligned} \quad (43)$$

$$\arccos(r, x) = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (44a)$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \quad (44b)$$

$$a_0 = x \quad (44c)$$

$$\begin{aligned} \cos: \{ (r, s) \in \mathbb{R}^2 \mid r > 0 \} &\rightarrow \mathbb{R} \\ (r, s) &\mapsto x \end{aligned} \quad (45)$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \quad (46a)$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \quad (46b)$$

$$a_0 = \cos(r, s) \quad (46c)$$

$$s = \lim_{m \rightarrow \infty} 2^m \sqrt{2r^2 - 2ra_{m,m}} \quad (47a)$$

$$a_{m,n} = \frac{\sqrt{2r^2 + 2ra_{m,n-1}}}{2} \quad \text{for } n \leq m \quad (47b)$$

$$a_{m,0} = \cos(r, s) \quad (47c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (48a)$$

$$a_{m,n-1} = \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \quad (48b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (48c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (49a)$$

$$2a_{m,n-1} = 2 \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \quad (49b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (49c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (50a)$$

$$2a_{m,n-1} = \frac{2\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \quad (50b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (50c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (51a)$$

$$2a_{m,n-1} = \sqrt{2r^2 + 2ra_{m,n}} \quad \text{for } n \leq m \quad (51b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (51c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (52a)$$

$$(2a_{m,n-1})^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2 \quad \text{for } n \leq m \quad (52b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (52c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (53a)$$

$$2^2 a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2 \quad \text{for } n \leq m \quad (53b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (53c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (54a)$$

$$4a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2 \quad \text{for } n \leq m \quad (54b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (54c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (55a)$$

$$4a_{m,n-1}^2 = 2r^2 + 2ra_{m,n} \quad \text{for } n \leq m \quad (55b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (55c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (56a)$$

$$4a_{m,n-1}^2 - 2r^2 = 2r^2 + 2ra_{m,n} - 2r^2 \quad \text{for } n \leq m \quad (56b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (56c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (57a)$$

$$4a_{m,n-1}^2 - 2r^2 = 2ra_{m,n} \quad \text{for } n \leq m \quad (57b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (57c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (58a)$$

$$\frac{4a_{m,n-1}^2 - 2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \quad (58b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (58c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (59a)$$

$$\frac{4a_{m,n-1}^2}{2r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \quad (59b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (59c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (60a)$$

$$\frac{2a_{m,n-1}^2}{r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \quad (60b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (60c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (61a)$$

$$\frac{2a_{m,n-1}^2}{r} - r = \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \quad (61b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (61c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (62a)$$

$$\frac{2a_{m,n-1}^2}{r} - r = a_{m,n} \quad \text{for } n \leq m \quad (62b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (62c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (63a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (63b)$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s \quad (63c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (64a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (64b)$$

$$\frac{2^m \sqrt{2r^2 - 2ra_{m,0}}}{2^m} = \frac{s}{2^m} \quad (64c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (65a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (65b)$$

$$\sqrt{2r^2 - 2ra_{m,0}} = \frac{s}{2^m} \quad (65c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (66a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (66b)$$

$$\left(\sqrt{2r^2 - 2ra_{m,0}} \right)^2 = \left(\frac{s}{2^m} \right)^2 \quad (66c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (67a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (67b)$$

$$2r^2 - 2ra_{m,0} = \left(\frac{s}{2^m} \right)^2 \quad (67c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (68a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (68b)$$

$$2r^2 - 2ra_{m,0} = \frac{s^2}{(2^m)^2} \quad (68c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (69a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (69b)$$

$$2r^2 - 2ra_{m,0} = \frac{s^2}{2^{2m}} \quad (69c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (70a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (70b)$$

$$2r^2 - 2ra_{m,0} - 2r^2 = \frac{s^2}{2^{2m}} - 2r^2 \quad (70c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (71a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (71b)$$

$$-2ra_{m,0} = \frac{s^2}{2^{2m}} - 2r^2 \quad (71c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (72a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (72b)$$

$$\frac{-2ra_{m,0}}{-2r} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r} \quad (72c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (73a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (73b)$$

$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r} \quad (73c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (74a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (74b)$$

$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right)}{-2r} - \frac{2r^2}{-2r} \quad (74c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (75a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (75b)$$

$$a_{m,0} = \frac{s^2}{2^{2m}(-2r)} - \frac{2r^2}{-2r} \quad (75c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (76a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (76b)$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} - \frac{2r^2}{-2r} \quad (76c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (77a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (77b)$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + \frac{-2r^2}{-2r} \quad (77c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (78a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (78b)$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + r \quad (78c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (79a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (79b)$$

$$a_{m,0} = r + \frac{s^2}{-2^{2m+1}r} \quad (79c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (80a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (80b)$$

$$a_{m,0} = r - \left(-\frac{s^2}{-2^{2m+1}r} \right) \quad (80c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (81a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (81b)$$

$$a_{m,0} = r - \left(\frac{-s^2}{-2^{2m+1}r} \right) \quad (81c)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (82a)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (82b)$$

$$a_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (82c)$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (83a)$$

$$a_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (83b)$$

$$r\hat{a}_{m,n} = \frac{2(r\hat{a}_{m,n-1})^2}{r} - r \quad \text{for } n \leq m \quad (84a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (84b)$$

$$r\hat{a}_{m,n} = \frac{2r^2\hat{a}_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \quad (85a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (85b)$$

$$r\hat{a}_{m,n} = 2r\hat{a}_{m,n-1}^2 - r \quad \text{for } n \leq m \quad (86a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (86b)$$

$$\frac{r\hat{a}_{m,n}}{r} = \frac{2r\hat{a}_{m,n-1}^2 - r}{r} \quad \text{for } n \leq m \quad (87a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (87b)$$

$$\hat{a}_{m,n} = \frac{2r\hat{a}_{m,n-1}^2 - r}{r} \quad \text{for } n \leq m \quad (88a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (88b)$$

$$\hat{a}_{m,n} = \frac{2r\hat{a}_{m,n-1}^2}{r} - \frac{r}{r} \quad \text{for } n \leq m \quad (89a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (89b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - \frac{r}{r} \quad \text{for } n \leq m \quad (90a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (90b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (91a)$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r} \quad (91b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (92a)$$

$$\frac{r\hat{a}_{m,0}}{r} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r} \quad (92b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (93a)$$

$$\hat{a}_{m,0} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r} \quad (93b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (94a)$$

$$\hat{a}_{m,0} = \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r} \quad (94b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (95a)$$

$$\hat{a}_{m,0} = 1 - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r} \quad (95b)$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m \quad (96a)$$

$$\hat{a}_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2} \quad (96b)$$

$$\hat{a}_{m,n} = T_2^{\circ n}(\hat{a}_{m,0}) \quad \text{for } n \leq m \quad (97)$$

$$\hat{a}_{m,n} = T_2^{\circ n} \left(1 - \frac{s^2}{2^{2m+1}r^2} \right) \quad \text{for } n \leq m \quad (98)$$

$$\hat{a}_{m,n} = T_{2^n} \left(1 - \frac{s^2}{2^{2m+1}r^2} \right) \quad \text{for } n \leq m \quad (99)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(\left(1 - \frac{s^2}{2^{2m+1}r^2} \right) - 1 \right)^k \quad \text{for } n \leq m \quad (100)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(1 - \frac{s^2}{2^{2m+1}r^2} - 1 \right)^k \quad \text{for } n \leq m \quad (101)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \quad \text{for } n \leq m \quad (102)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \left(\frac{2^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \right) \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \quad \text{for } n \leq m \quad (103)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (104)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(2 \left(-\frac{s^2}{2^{2m+1}r^2} \right) \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (105)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-2 \left(\frac{s^2}{2^{2m+1}r^2} \right) \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (106)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{2s^2}{2^{2m+1}r^2} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (107)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{2^{2m}r^2} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (108)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{(2^2)^m r^2} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (109)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{4^m r^2} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (110)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s^2}{4^m r^2} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (111)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s}{r} \right)^2}{4^m} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (112)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s}{r} \right)^2}{4^m} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (113)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s}{r} \right)^2}{(4^m)^k} \right)^k}{(2k)!} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (114)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2) \quad \text{for } n \leq m \quad (115)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \frac{\prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k} \quad \text{for } n \leq m \quad (116)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \frac{\prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{\prod_{j=0}^{k-1} 4^m} \quad \text{for } n \leq m \quad (117)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m} \quad \text{for } n \leq m \quad (118)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m} \quad \text{for } n \leq m \quad (119)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{4^n - j^2}{4^m} \quad \text{for } n \leq m \quad (120)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(\frac{4^n}{4^m} - \frac{j^2}{4^m} \right) \quad \text{for } n \leq m \quad (121)$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right) \quad \text{for } n \leq m \quad (122)$$

$$r \hat{a}_{m,n} = r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right) \quad \text{for } n \leq m \quad (123)$$

$$a_{m,n} = r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right) \quad \text{for } n \leq m \quad (124)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \quad (125)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{m-m} - \frac{j^2}{4^m}\right) \quad (126)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^0 - \frac{j^2}{4^m}\right) \quad (127)$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \quad (128)$$

$$\cos(r, s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \quad (129)$$

$$\begin{aligned} \arcsin: \{(r, y) \in \mathbb{R}^2 \mid r > 0, -r \leq y \leq r\} &\rightarrow \mathbb{R} \\ (r, y) &\mapsto s \end{aligned} \quad (130)$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, y)}{2} \quad (131)$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, y) + \arccos(r, y) - \arccos(r, y)}{2} \quad (132)$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, -y) + \arccos(r, -y) - \arccos(r, y)}{2} \quad (133)$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) - \arcsin(r, y) + \arccos(r, -y) - \arccos(r, y)}{2} \quad (134)$$

$$\arcsin(r, y) = \frac{\arccos(r, -y) - \arccos(r, y)}{2} \quad (135)$$

$$\begin{aligned} \sin: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} &\rightarrow \mathbb{R} \\ (r, s) &\mapsto y \end{aligned} \quad (136)$$

$$\sin(r, s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \quad (137)$$

$$\cos(r, \alpha - \beta) = a_{0,0} \quad (138a)$$

$$a_{0,0} = r - \frac{\left(\sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2} \right)^2}{2r} \quad (138b)$$

$$\cos(r, \alpha - \beta) = r - \frac{\left(\sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2} \right)^2}{2r} \quad (139)$$

$$\cos(r, \alpha - \beta) = r - \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}{2r} \quad (140)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2}{2r} - \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}{2r} \quad (141)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - ((\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2)}{2r} \quad (142)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos(r, \alpha) - \cos(r, \beta))^2 - (\sin(r, \alpha) - \sin(r, \beta))^2}{2r} \quad (143)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta)) - (\sin(r, \alpha) - \sin(r, \beta))^2}{2r} \quad (144)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta)) - (\sin^2(r, \alpha) - 2\sin(r, \alpha)\sin(r, \beta) + \sin^2(r, \beta))}{2r} \quad (145)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - \cos^2(r, \alpha) + 2\cos(r, \alpha)\cos(r, \beta) - \cos^2(r, \beta) - (\sin^2(r, \alpha) - 2\sin(r, \alpha)\sin(r, \beta) + \sin^2(r, \beta))}{2r} \quad (146)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - \cos^2(r, \alpha) + 2\cos(r, \alpha)\cos(r, \beta) - \cos^2(r, \beta) - \sin^2(r, \alpha) + 2\sin(r, \alpha)\sin(r, \beta) - \sin^2(r, \beta)}{2r} \quad (147)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) + \sin^2(r, \alpha)) + 2\cos(r, \alpha)\cos(r, \beta) - \cos^2(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta) - \sin^2(r, \beta)}{2r} \quad (148)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) + \sin^2(r, \alpha)) - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2 \cos(r, \alpha) \cos(r, \beta) + 2 \sin(r, \alpha) \sin(r, \beta)}{2r} \quad (149)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) + \sin^2(r, \alpha)) - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2 (\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta))}{2r} \quad (150)$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - r^2 - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2 (\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta))}{2r} \quad (151)$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2 (\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta))}{2r} \quad (152)$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - r^2 + 2 (\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta))}{2r} \quad (153)$$

$$\cos(r, \alpha - \beta) = \frac{2 (\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta))}{2r} \quad (154)$$

$$\cos(r, \alpha - \beta) = \frac{\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta)}{r} \quad (155)$$

$$\cos(r, \alpha \pm \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha \pm \beta}{r}\right)^{2k}}{(2k)!} \quad (156)$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha \pm \beta}{r}\right)^{2k}}{(2k)!} \quad (157)$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} \binom{2k}{j} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{\pm \beta}{r}\right)^j}{(2k)!} \quad (158)$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} \frac{(2k)!}{j!(2k-j)!} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{\pm \beta}{r}\right)^j}{(2k)!} \quad (159)$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} (2k)! \frac{(\frac{\alpha}{r})^{2k-j} (\frac{\pm\beta}{r})^j}{j!(2k-j)!}}{(2k)!} \quad (160)$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k (2k)! \sum_{j=0}^{2k} \frac{(\frac{\alpha}{r})^{2k-j} (\frac{\pm\beta}{r})^j}{j!(2k-j)!}}{(2k)!} \quad (161)$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n (-1)^k \sum_{j=0}^{2k} \frac{(\frac{\alpha}{r})^{2k-j} (\frac{\pm\beta}{r})^j}{j!(2k-j)!} \quad (162)$$

$$\cos(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j (\frac{\beta}{r})^{2j}}{(2j)!} \mp \sum_{k=0}^{n-1} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j (\frac{\beta}{r})^{2j+1}}{(2j+1)!} \right) \quad (163)$$

$$\cos(r, \alpha \pm \beta) = r \left(\lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j (\frac{\beta}{r})^{2j}}{(2j)!} \right) \mp \lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j (\frac{\beta}{r})^{2j+1}}{(2j+1)!} \right) \right) \quad (164)$$

$$\cos(r, \alpha \pm \beta) = r \left(\sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} \mp \lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j (\frac{\beta}{r})^{2j+1}}{(2j+1)!} \right) \right) \quad (165)$$

$$\cos(r, \alpha \pm \beta) = r \left(\sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} \mp \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!} \right) \quad (166)$$

$$\cos(r, \alpha \pm \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} \mp r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!} \quad (167)$$

$$\cos(r, \alpha \pm \beta) = \frac{r \left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} \mp r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!} \right)}{r} \quad (168)$$

$$\cos(r, \alpha \pm \beta) = \frac{r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \mp r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \quad (169)$$

$$\cos(r, \alpha \pm \beta) = \frac{\left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right) \mp r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \quad (170)$$

$$\cos(r, \alpha \pm \beta) = \frac{\left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right) \mp \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \quad (171)$$

$$\cos(r, \alpha \pm \beta) = \frac{\cos(r, \alpha) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right) \mp \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \quad (172)$$

$$\cos(r, \alpha \pm \beta) = \frac{\cos(r, \alpha) \cos(r, \beta) \mp \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \quad (173)$$

$$\sin(r, \alpha) \sin(r, \beta) = \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right) \quad (174)$$

$$\sin(r, \alpha) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \quad (175)$$

$$\sin(r, \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \quad (176)$$

$$\cos(r, \alpha \pm \beta) = \frac{\cos(r, \alpha) \cos(r, \beta) \mp \sin(r, \alpha) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \quad (177)$$

$$\cos(r, \alpha \pm \beta) = \frac{\cos(r, \alpha) \cos(r, \beta) \mp \sin(r, \alpha) \sin(r, \beta)}{r} \quad (178)$$

$$\sin(r, \cdot) : s \mapsto r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \quad (179)$$

$$\sin(r, \alpha \pm \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha \pm \beta}{r}\right)^{2k+1}}{(2k+1)!} \quad (180)$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha \pm \beta}{r}\right)^{2k+1}}{(2k+1)!} \quad (181)$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k+1} \binom{2k+1}{j} \left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm \beta}{r}\right)^j}{(2k+1)!} \quad (182)$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k+1} \frac{(2k+1)!}{j!(2k+1-j)!} \left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm \beta}{r}\right)^j}{(2k+1)!} \quad (183)$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k+1} (2k+1)! \frac{\left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm \beta}{r}\right)^j}{j!(2k+1-j)!}}{(2k+1)!} \quad (184)$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k (2k+1)! \sum_{j=0}^{2k+1} \frac{\left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm \beta}{r}\right)^j}{j!(2k+1-j)!}}{(2k+1)!} \quad (185)$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n (-1)^k \sum_{j=0}^{2k+1} \frac{\left(\frac{\alpha}{r}\right)^{2k+1-j} \left(\frac{\pm \beta}{r}\right)^j}{j!(2k+1-j)!} \quad (186)$$

$$\sin(r, \alpha \pm \beta) = r \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j}}{(2j)!} \pm \sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \quad (187)$$

$$\sin(r, \alpha \pm \beta) = r \left(\lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j}}{(2j)!} \right) \pm \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \right) \quad (188)$$

$$\sin(r, \alpha \pm \beta) = r \left(\sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \pm \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j \left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!} \right) \right) \quad (189)$$

$$\sin(r, \alpha \pm \beta) = r \left(\sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \pm \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right) \quad (190)$$

$$\sin(r, \alpha \pm \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \pm r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \quad (191)$$

$$\sin(r, \alpha \pm \beta) = \frac{r \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \pm r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (192)$$

$$\sin(r, \alpha \pm \beta) = \frac{r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \pm r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \quad (193)$$

$$\sin(r, \alpha \pm \beta) = \frac{\left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \right) \pm r^2 \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \quad (194)$$

$$\sin(r, \alpha \pm \beta) = \frac{\left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \right) \pm \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (195)$$

$$\sin(r, \alpha \pm \beta) = \frac{\sin(r, \alpha) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k}}{(2k)!} \right) \pm \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (196)$$

$$\sin(r, \alpha \pm \beta) = \frac{\sin(r, \alpha) \cos(r, \beta) \pm \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r}\right)^{2k}}{(2k)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (197)$$

$$\sin(r, \alpha \pm \beta) = \frac{\sin(r, \alpha) \cos(r, \beta) \pm \cos(r, \alpha) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r} \right)^{2k+1}}{(2k+1)!} \right)}{r} \quad (198)$$

$$\sin(r, \alpha \pm \beta) = \frac{\sin(r, \alpha) \cos(r, \beta) \pm \cos(r, \alpha) \sin(r, \beta)}{r} \quad (199)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \right)}{\partial s} \quad (200)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k}}{r^{2k}}}{(2k)!} \right)}{\partial s} \quad (201)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \quad (202)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \quad (203)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k-1} (2k)!} \right)}{\partial s} \quad (204)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k}}{\partial s}}{r^{2k-1} (2k)!} \quad (205)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) s^{2k-1}}{r^{2k-1} (2k)!} \quad (206)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \frac{s^{2k-1}}{r^{2k-1}}}{(2k)!} \quad (207)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)!} \quad (208)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)(2k-1)!} \quad (209)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \quad (210)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{-(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \quad (211)$$

$$\frac{\partial \cos(r, s)}{\partial s} = - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \quad (212)$$

$$\frac{\partial \cos(r, s)}{\partial s} = - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2k-1)!} \quad (213)$$

$$\frac{\partial \cos(r, s)}{\partial s} = - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2(k-1)+1)!} \quad (214)$$

$$\frac{\partial \cos(r, s)}{\partial s} = - \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \quad (215)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{-r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!}}{r} \quad (216)$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{-\sin(r, s)}{r} \quad (217)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \right)}{\partial s} \quad (218)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k+1}}{r^{2k+1}}}{(2k+1)!} \right)}{\partial s} \quad (219)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k+1}(2k+1)!} \right)}{\partial s} \quad (220)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k+1}}{r^{2k+1}(2k+1)!} \right)}{\partial s} \quad (221)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k}(2k+1)!} \right)}{\partial s} \quad (222)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k+1}}{\partial s}}{r^{2k}(2k+1)!} \quad (223)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) s^{2k}}{r^{2k}(2k+1)!} \quad (224)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \frac{s^{2k}}{r^{2k}}}{(2k+1)!} \quad (225)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)!} \quad (226)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)(2k)!} \quad (227)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \quad (228)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}}{r} \quad (229)$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\cos(r, s)}{r} \quad (230)$$

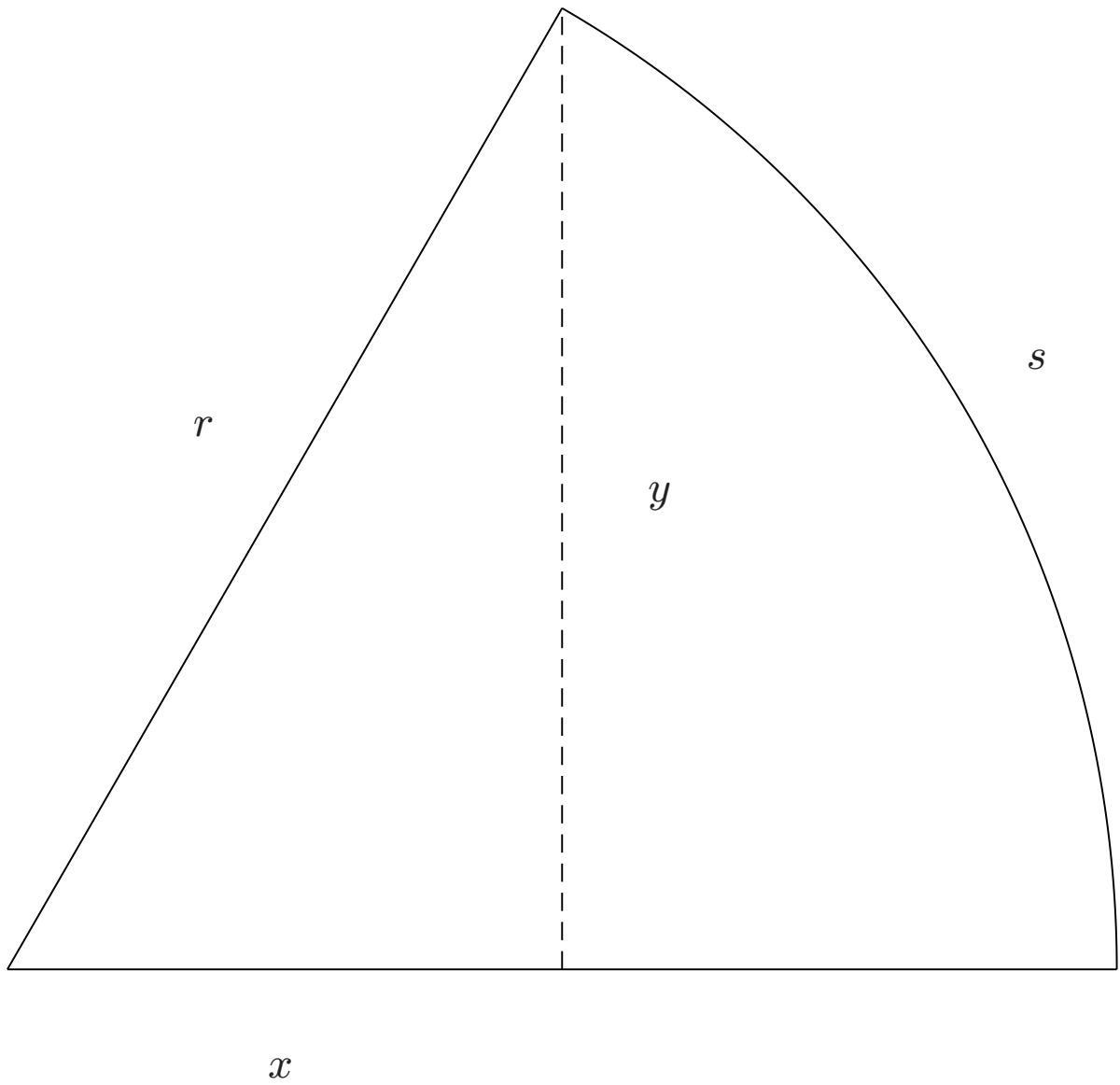


Figure 1: Circular sector.

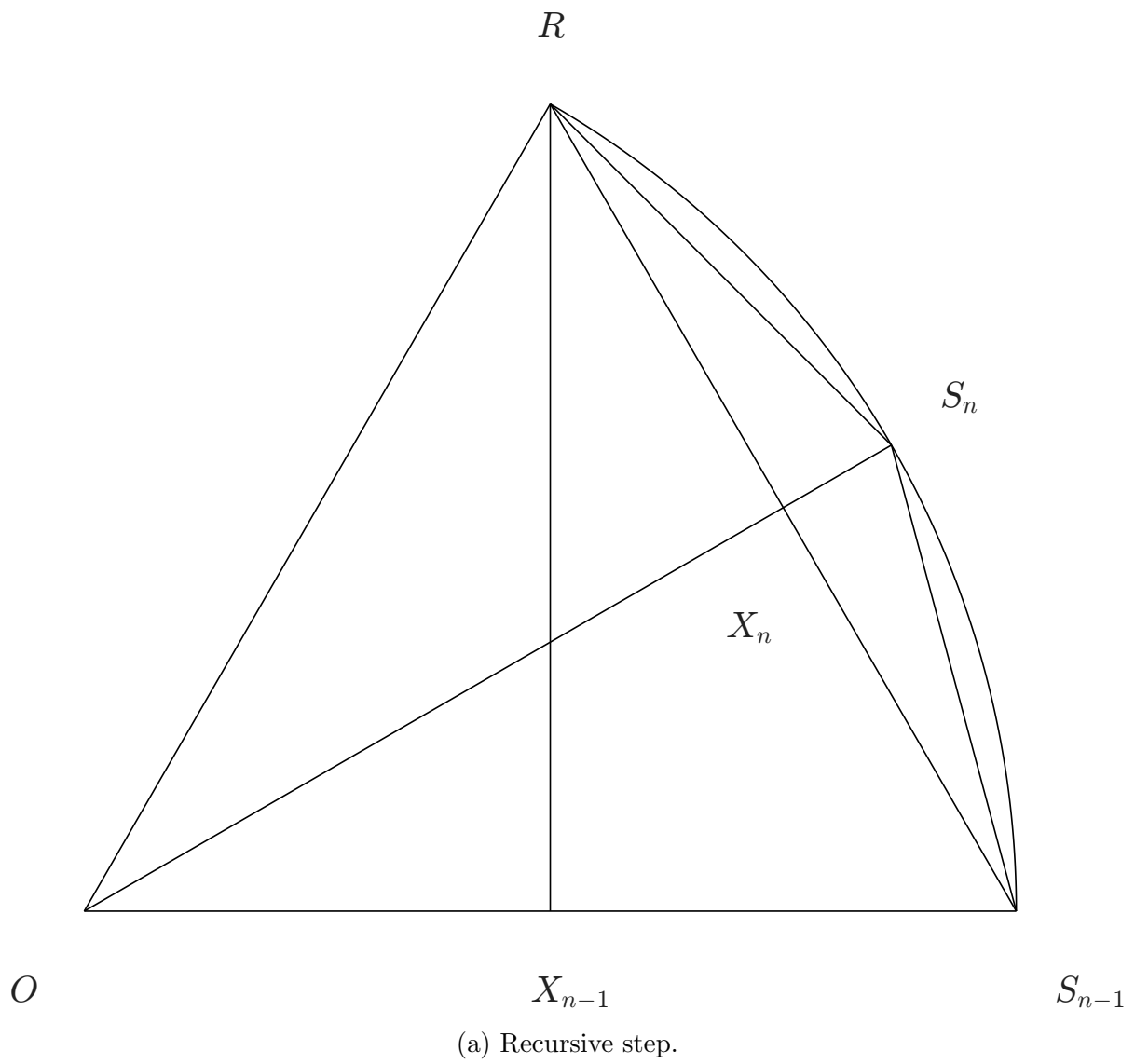


Figure 2: Circular sector sequence.

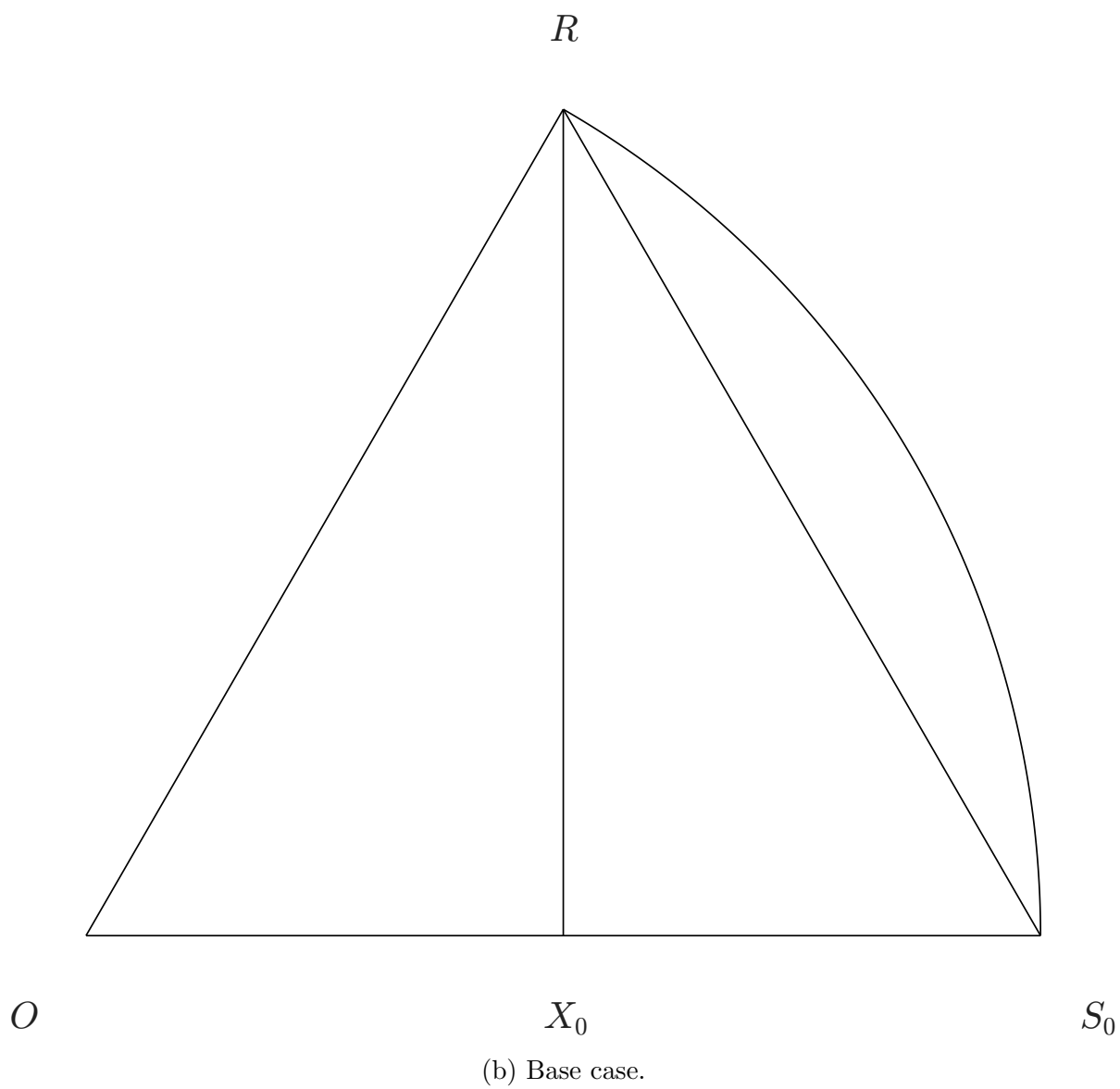


Figure 2: Circular sector sequence.

Simple harmonic motion

$$F = ma = -kx \quad (231)$$

$$ma = -kx \quad (232)$$

$$x(t) = \cos(r, \omega t) \quad (233)$$

$$a(t) = \frac{d^2 x}{dt^2} \quad (234)$$

$$a(t) = \frac{d^2 \cos(r, \omega t)}{dt^2} \quad (235)$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2} \quad (236)$$

$$a(x) = \frac{-\omega^2 x}{r^2} \quad (237)$$

$$m \frac{-\omega^2 x}{r^2} = -kx \quad (238)$$

$$\frac{m(-\omega^2 x)}{r^2} = -kx \quad (239)$$

$$\frac{-m\omega^2 x}{r^2} = -kx \quad (240)$$

$$\frac{-m\omega^2 x}{r^2} = \frac{-kr^2 x}{r^2} \quad (241)$$

$$m\omega^2 = kr^2 \quad (242)$$

$$v(t) = \frac{dx}{dt} \quad (243)$$

$$v(t) = \frac{d \cos(r, \omega t)}{dt} \quad (244)$$

$$v(t) = \frac{-\omega \sin(r, \omega t)}{r} \quad (245)$$

$$K(t) = \frac{mv^2(t)}{2} \quad (246)$$

$$K(t) = \frac{m \left(\frac{-\omega \sin(r, \omega t)}{r} \right)^2}{2} \quad (247)$$

$$K(t) = \frac{m \frac{(-\omega \sin(r, \omega t))^2}{r^2}}{2} \quad (248)$$

$$K(t) = \frac{m \frac{(-\omega)^2 \sin^2(r, \omega t)}{r^2}}{2} \quad (249)$$

$$K(t) = \frac{m \frac{\omega^2 \sin^2(r, \omega t)}{r^2}}{2} \quad (250)$$

$$K(t) = \frac{\frac{m\omega^2 \sin^2(r, \omega t)}{r^2}}{2} \quad (251)$$

$$K(t) = \frac{\frac{kr^2 \sin^2(r, \omega t)}{r^2}}{2} \quad (252)$$

$$K(t) = \frac{k \sin^2(r, \omega t)}{2} \quad (253)$$

$$U(t) = \frac{kx^2(t)}{2} \quad (254)$$

$$U(t) = \frac{k \cos^2(r, \omega t)}{2} \quad (255)$$

$$E = K + U \quad (256)$$

$$E = \frac{k \sin^2(r, \omega t)}{2} + \frac{k \cos^2(r, \omega t)}{2} \quad (257)$$

$$E = \frac{k \sin^2(r, \omega t) + k \cos^2(r, \omega t)}{2} \quad (258)$$

$$E = \frac{k(\sin^2(r, \omega t) + \cos^2(r, \omega t))}{2} \quad (259)$$

$$E = \frac{kr^2}{2} \quad (260)$$

$$E = \frac{m\omega^2}{2} \quad (261)$$

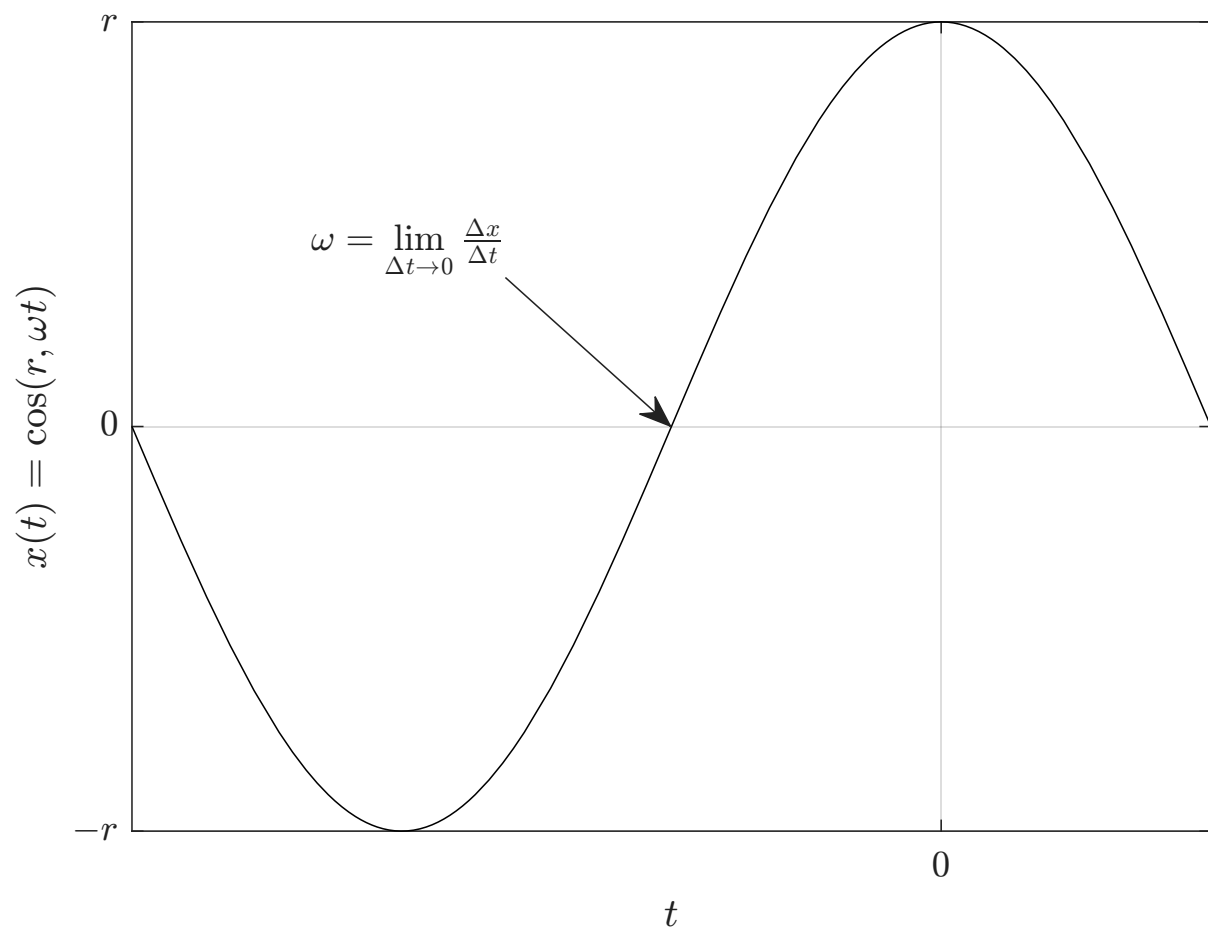


Figure 3: Simple harmonic motion.

Uniform circular motion

$$F = ma \quad (262)$$

$$\omega = \frac{\Delta s}{\Delta t} \quad (263)$$

$$\vec{r}_x(t) = \cos(r, \omega t) \quad (264)$$

$$\vec{r}_y(t) = \sin(r, \omega t) \quad (265)$$

$$\vec{r} = (\vec{r}_x, \vec{r}_y) \quad (266)$$

$$\vec{r}(t) = (\cos(r, \omega t), \sin(r, \omega t)) \quad (267)$$

$$\vec{v} = \frac{d\vec{r}}{dt} \quad (268)$$

$$\vec{v}(t) = \frac{d(\cos(r, \omega t), \sin(r, \omega t))}{dt} \quad (269)$$

$$\vec{v}(t) = \left(\frac{d \cos(r, \omega t)}{dt}, \frac{d \sin(r, \omega t)}{dt} \right) \quad (270)$$

$$\vec{v}(t) = \left(\frac{-\omega \sin(r, \omega t)}{r}, \frac{\omega \cos(r, \omega t)}{r} \right) \quad (271)$$

$$\vec{v}(t) = \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r} \quad (272)$$

$$\vec{a} = \frac{d^2 \vec{r}}{dt^2} \quad (273)$$

$$\vec{a}(t) = \frac{d^2(\cos(r, \omega t), \sin(r, \omega t))}{dt^2} \quad (274)$$

$$\vec{a}(t) = \left(\frac{d^2 \cos(r, \omega t)}{dt^2}, \frac{d^2 \sin(r, \omega t)}{dt^2} \right) \quad (275)$$

$$\vec{a}(t) = \left(\frac{-\omega^2 \cos(r, \omega t)}{r^2}, \frac{-\omega^2 \sin(r, \omega t)}{r^2} \right) \quad (276)$$

$$\vec{a}(t) = \frac{-\omega^2 (\cos(r, \omega t), \sin(r, \omega t))}{r^2} \quad (277)$$

$$\vec{a} = \frac{-\omega^2 \vec{r}}{r^2} \quad (278)$$

$$\vec{F} = m\vec{a} \quad (279)$$

$$\vec{F} = m \frac{-\omega^2 \vec{r}}{r^2} \quad (280)$$

$$\vec{F} = \frac{m(-\omega^2 \vec{r})}{r^2} \quad (281)$$

$$\vec{F} = \frac{-m\omega^2 \vec{r}}{r^2} \quad (282)$$

$$v = |\vec{v}| \quad (283)$$

$$v = \left| \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r} \right| \quad (284)$$

$$v = \frac{|\omega (-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \quad (285)$$

$$v = \frac{|\omega| |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \quad (286)$$

$$v = \frac{\omega |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \quad (287)$$

$$v = \frac{\omega \sqrt{(-\sin(r, \omega t))^2 + \cos^2(r, \omega t)}}{|r|} \quad (288)$$

$$v = \frac{\omega \sqrt{\sin^2(r, \omega t) + \cos^2(r, \omega t)}}{|r|} \quad (289)$$

$$v = \frac{\omega \sqrt{r^2}}{|r|} \quad (290)$$

$$v = \frac{\omega |r|}{|r|} \quad (291)$$

$$v = \omega \quad (292)$$

$$E = \frac{mv^2}{2} \quad (293)$$

$$E = \frac{m\omega^2}{2} \quad (294)$$

$$a = |\vec{a}| \quad (295)$$

$$a = \left| \frac{-\omega^2 \vec{r}}{r^2} \right| \quad (296)$$

$$a = \frac{|-\omega^2 \vec{r}|}{|r^2|} \quad (297)$$

$$a = \frac{|-\omega^2| |\vec{r}|}{|r^2|} \quad (298)$$

$$a = \frac{\omega^2 |\vec{r}|}{|r^2|} \quad (299)$$

$$a = \frac{\omega^2 r}{|r^2|} \quad (300)$$

$$a = \frac{\omega^2 r}{r^2} \quad (301)$$

$$a = \frac{\omega^2}{r} \quad (302)$$

$$F = \frac{m\omega^2}{r} \quad (303)$$

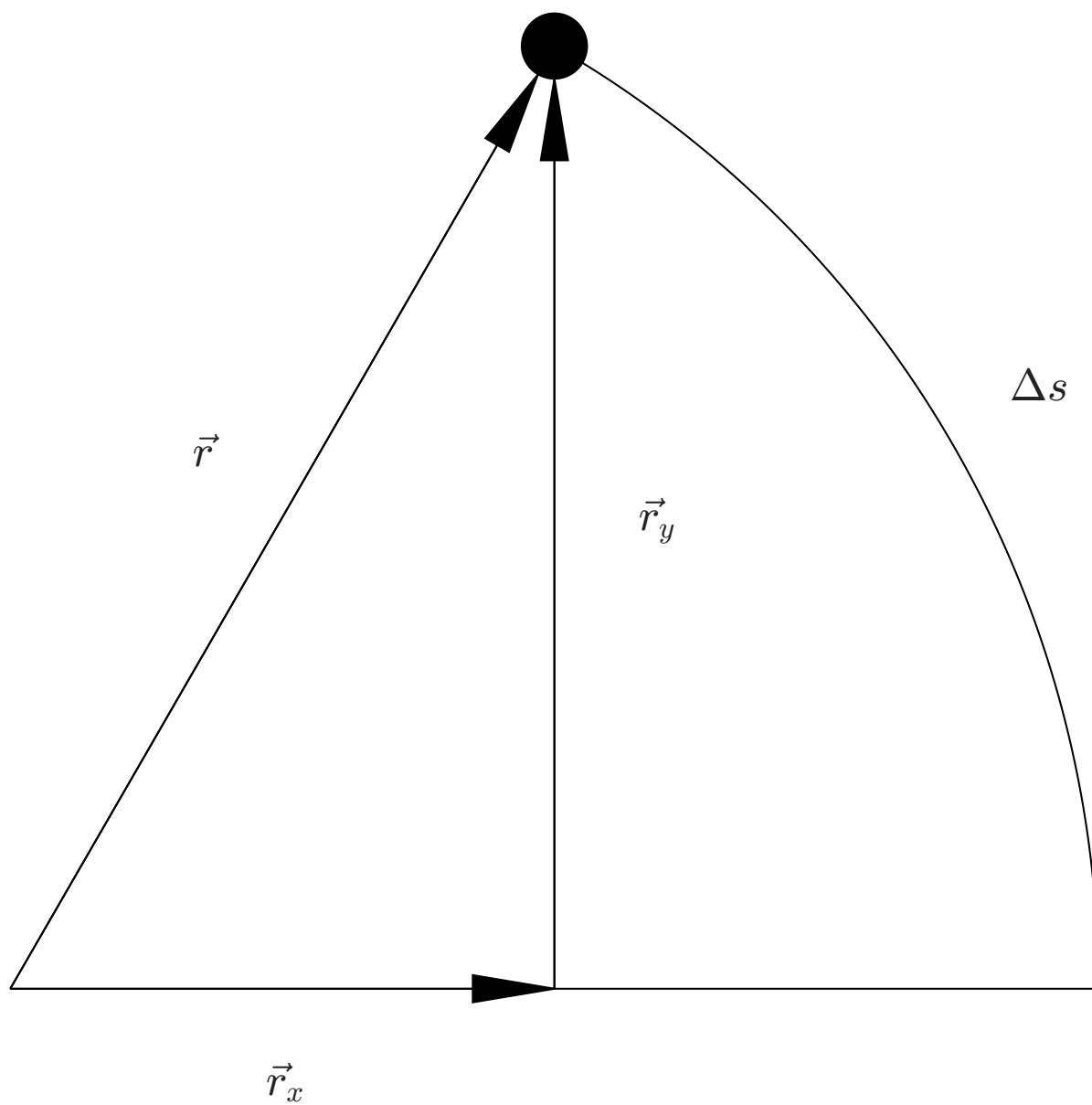


Figure 4: Uniform circular motion.